Elastic Collision Conservation Derivation

A proton of mass m enters with speed v_1. After an elastic collision with a stationary particle of mass 4m, the proton leaves at 72 degrees with speed v_3, while the 4m particle recoils at angle theta with speed v_4.

1. Original conservation equations

x momentum	$m v_1 = m v_3 \cos(72 \text{ degrees}) + 4m v_4 \cos(\text{theta})$
y momentum	$0 = m v_3 \sin(72 \text{ degrees}) - 4m v_4 \sin(\text{theta})$
kinetic energy	$1/2\ m v_1^2 = 1/2\ m v_3^2 + 1/2(4m) v_4^2$

The factors m and 4m are retained explicitly above, exactly as in the class setup. The target is initially at rest, so it contributes neither initial momentum nor initial kinetic energy.

2. Make the system dimensionless

Define the speed ratios $x = v_3/v_1$ and $y = v_4/v_1$. Divide momentum by $m v_1$ and energy by $(1/2)m v_1^2$:

$$1 = x \cos 72 \text{ degrees} + 4y \cos \text{theta}$$

$$0 = x \sin 72 \text{ degrees} - 4y \sin \text{theta}$$

$$1 = x^2 + 4y^2$$

3. Eliminate the recoil angle

Rearrange the two momentum components:

$$4y \cos \text{theta} = 1 - x \cos 72 \text{ degrees}$$

$$4y \sin \text{theta} = x \sin 72 \text{ degrees}$$

Square and add. Since $\cos^2 \text{theta} + \sin^2 \text{theta} = 1$:

$$16y^2 = (1 - x \cos 72 \text{ degrees})^2 + (x \sin 72 \text{ degrees})^2$$

$$16y^2 = 1 - 2x \cos 72 \text{ degrees} + x^2.$$

From energy conservation, $4y^2 = 1 - x^2$, so $16y^2 = 4(1 - x^2)$. Equating the two expressions gives

$$\mathbf{5x^2 - 2x \cos 72 \text{ degrees} - 3 = 0.}$$

4. Solve for the outgoing speeds and angle

Applying the quadratic formula:

$$x = [\cos 72 \text{ degrees} +/- \sqrt{(\cos^2 72 \text{ degrees} + 15)}] / 5$$

$$x = 0.838861737 \text{ (physical positive-speed branch)}$$

The other algebraic root is x = -0.715254939. With v_3 defined as a nonnegative speed along the stated 72-degree direction, that root is not the stated physical geometry.

Now use energy conservation:

$$y = (1/2)\sqrt{(1 - x^2)} = 0.272172274$$

Therefore $v_3 = 0.838861737\ v_1$ and $v_4 = 0.272172274\ v_1$.

The recoil angle follows from the component equations:

$$\tan \theta = [x \sin 72 \text{ degrees}] / [1 - x \cos 72 \text{ degrees}]$$

theta = 47.122693 degrees.

5. Scattered proton energy fraction

The scattered proton is the outgoing mass-m particle, so its kinetic-energy fraction is

$$K_3/K_1 = [(1/2)m\ v_3^2] / [(1/2)m\ v_1^2] = x^2$$

$K_3/K_1 = 0.703689013 = 70.3689\%$.

The recoil particle receives the remaining 29.6311%: $K_4/K_1 = 4y^2 = 0.296310987$.

6. Numerical verification

quantity	evaluated result	expected
x momentum	1.000000000000	1
y momentum	1.11e-16	0
kinetic energy	1.000000000000	1
energy fractions	0.703689013 + 0.296310987	1.000000000

All checks close to rounding precision. The previously vulnerable arithmetic is therefore independently confirmed.